

Brief Introduction

The increasing cost of purchasing equipment such as speakers, projectors, microphones, cameras, extension boards, gaming consoles, and event accessories often makes ownership impractical for people who require these items only occasionally. At the same time, many individuals own such equipment that remains unused for a significant portion of the time.

RentANY is a web-based peer-to-peer rental platform that enables users to rent and lend items on an hourly basis. The scope of the platform is limited to permissible, non-restricted items only; prohibited or regulated goods are outside the scope of this system. Owners can list their items by providing images, descriptions, rental charges, and availability, while renters can browse available items, view details, and request bookings.

The platform aims to create a secure and user-friendly rental ecosystem where individuals can earn passive income by renting out their unused belongings while providing affordable access to equipment for others.

The system will include user authentication, item management, booking requests, availability management, payment and security deposit handling, reviews and ratings, and user dashboards for managing rentals and listings.

Aim

The primary objectives of the project are:

- To develop an online platform for renting items on an hourly basis.
- To allow users to list their own items with images, descriptions, rental charges, and availability.
- To provide a simple and secure booking and payment system between renters and owners.
- To maintain transparency and accountability through user profiles, ratings, and reviews.
- To simplify the process of renting equipment while enabling owners to generate income from underutilized assets.
- To provide real-time map-based route guidance between the renter and the item provider for convenient pickup and drop-off.
- To integrate an AI-powered chatbot that recommends the items required for a described occasion, helping users quickly find and rent everything they need.

Tools to be Used

Frontend: React.js, Tailwind CSS, HTML5, CSS3, JavaScript

Backend: Node.js, Express.js

Database: PostgreSQL

ORM: Prisma ORM

Authentication: JWT (JSON Web Token), bcrypt

Payments: Razorpay / Stripe (test mode) — for platform fee and security deposit transactions

Image Storage: Cloudinary

Development Tools: Visual Studio Code, Git & GitHub, Postman

Deployment: Vercel (Frontend & Backend)

Maps & Navigation: Google Maps API / Mapbox — for live location and route mapping between renter and provider

AI Integration: Grok API (xAI) — for the occasion-based smart rental recommendation chatbot

Expected Outcome

At the completion of the project, the system is expected to:

- Provide a responsive web application for renting various items on an hourly basis.
- Allow users to register, log in, and manage their profiles.
- Enable owners to upload and manage rental listings with images and pricing.
- Allow renters to search, filter, and request items based on availability.
- Charge a fixed platform fee on each booking as the platform's revenue model.
- Collect a refundable security deposit at the time of booking, to be withheld (fully or partially) in case of item damage or usage beyond the rented duration, and refunded to the renter otherwise.
- Provide an admin panel for basic moderation of listings, users, and dispute cases (e.g., damage claims, no-shows, late returns).
- Maintain booking history for both owners and renters.
- Provide ratings and reviews to improve trust between users.
- Offer an intuitive dashboard for managing listings, bookings, payments, and deposits.
- Display an interactive map with the route between renter and provider for easy pickup and drop-off.
- Provide an AI chatbot that, given a described occasion, suggests and returns a list of the items required for it.

The final application will demonstrate practical implementation of full-stack web development, database management, authentication, payment integration, RESTful APIs, and cloud deployment.

Gantt Chart

